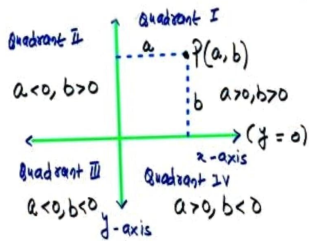


3-D Geometry

Introduction



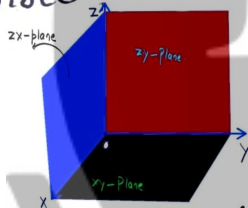
* **x-coordinate (abscissa)**

• Represents the distance from y-axis

* **y-coordinate (ordinate)**

• Represent the distance from x-axis

* In a 3D-space, to represent a point we need a set of three numbers.

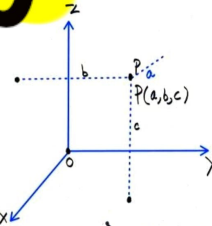


* **x-coordinate:**
Distance of a point from yz-plane

* **y-coordinate:**
Distance of a point from xz-plane

* **z-coordinate:**
Distance of a point from xy-plane

* xy-plane, yz-plane, zx-plane divides the space into 8-parts, each part is known as an octant.



- Equation of x-axis: $y=0, z=0$
- Equation of y-axis: $x=0, z=0$
- Equation of z-axis: $x=0, y=0$

Sign of co-ordinates of a point in octants

Octants Coordinate	I	II	III	IV	V	VI	VII	VIII
x	+	-	-	+	+	-	-	+
y	+	+	-	-	+	+	-	-
z	+	+	+	+	-	-	-	-

Distance between Two points

(x_1, y_1, z_1) A $B(x_2, y_2, z_2)$

Distance b/w A (x_1, y_1, z_1) and B (x_2, y_2, z_2)

is given by

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example

Find the distance between the points P(1, -3, 4) and Q(-4, 1, 2)

(Sol) By distance formula

$$PQ = \sqrt{(-4-1)^2 + (1+3)^2 + (2-4)^2}$$

$$\Rightarrow PQ = \sqrt{25 + 16 + 4}$$

$$\Rightarrow PQ = \sqrt{45} = \sqrt{9 \times 5}$$

$$\Rightarrow PQ = 3\sqrt{5} \text{ units}$$

Example

Show that the points
 $P(-2, 3, 5)$, $Q(1, 2, 3)$ and
 $R(7, 0, -1)$ are collinear

Proof By distance formula

$$PQ = \sqrt{(1+2)^2 + (2-3)^2 + (3-5)^2}$$

$$PQ = \sqrt{9+1+4} = \sqrt{14} \text{ units}$$

$$QR = \sqrt{(7-1)^2 + (0-2)^2 + (-1-3)^2}$$

$$QR = \sqrt{36+4+16} = \sqrt{56}$$

$$\Rightarrow QR = \sqrt{4 \times 14}$$

$$QR = 2\sqrt{14} \text{ units}$$

$$\text{Also, } PR = \sqrt{(7+2)^2 + (0-3)^2 + (-1-5)^2}$$

$$= PR = \sqrt{81+9+36}$$

$$PR = \sqrt{126} = \sqrt{9 \times 14}$$

$$PR = 3\sqrt{14} \text{ unit}$$

$$\text{the clearly, } PQ + QR = \sqrt{14} + 2\sqrt{14}$$

$$= 3\sqrt{14}$$

$$\Rightarrow PQ + QR = PR$$

$\Rightarrow P, Q$ and R collinear points

Example

Find the point on x -axis which
is equidistant from the
point $A(3, 2, 2)$ and $B(5, 5, 4)$

(Sol) let the required
point be $(a, 0, 0)$

Such that $AC = BC$

$$\Rightarrow \sqrt{(a-3)^2 + 4 + 4} = \sqrt{(a-5)^2 + 25 + 16}$$

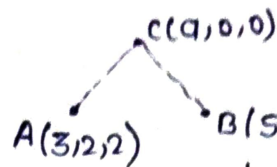
$$\Rightarrow a^2 + 9 - 6a + 8 = a^2 + 25 - 10a + 41$$

$$\Rightarrow 17 - 6a = 66 - 10a$$

$$\Rightarrow -60 + 10a = 66 - 17$$

$$\Rightarrow 4a = 49$$

$$\Rightarrow a = \frac{49}{4}$$



So that required point is
 $(\frac{49}{4}, 0, 0)$

Question (Ncert Example)

Distance of the point $(3, 4, 5)$
from the origin $(0, 0, 0)$ is

$$\sqrt{3^2 + 4^2 + 5^2}$$

$$= \sqrt{9 + 16 + 25}$$

$$= \sqrt{50}$$

Question (Ncert Example)

If the distance between the
points $(a, 0, 1)$ and $(0, 1, 2)$

is $\sqrt{27}$, then the value of a is

$$\sqrt{a^2 + 1} = \sqrt{27}$$

$$\Rightarrow a^2 + 1 = 27$$

$$\Rightarrow a^2 = 26$$

$$\Rightarrow a = \pm\sqrt{26}$$

$$\Rightarrow a = \pm 5$$

Question (Ncert Example)

The distance of point $P(3, 4, 5)$
from the yz -plane is

① 3 unit

② 4 unit

③ 5 unit

④ 500

• Distance from
 yz -plane

$\rightarrow x$ -co-ordinate

Question (Ncert Example)

The point $(-2, -3, -4)$ lies in the

① First octant

② Seventh octant

③ Second octant

④ Eighth octant

Question (Ncert Exemplar)
x-axis is the intersection of two planes

- ① xy and xz
- ② zy and zx
- ③ xy and yz
- ④ None of these

Question (Ncert Exemplar)

The point of y-axis is considered as $z=0, x=0$

- ① $x=0, y=0$
- ② $y=0, z=0$
- ✓ ③ $z=0, x=0$
- ④ None of these

Question (Ncert Exemplar)

A plane is parallel to yz-plane so it is perpendicular to

- ✓ ① x-axis
- ② y-axis
- ③ z-axis
- ④ None of these

Question (Ncert Exemplar)

The locus of a point for which $y=0, z=0$ is

- ✓ ① Equation of x-axis
- ② Equation of y-axis
- ③ Equation of z-axis
- ④ None of these

A point is on the xz plane
What are its y-coordinates

(Sol) y-coordination = 0

- * $x=0$ → equation of yz-plane
- * $y=0$ → equation of xz-plane
- * $z=0$ → equation of xy-plane